NBA non-contact injury survival — Update 3: neural survival models

Validation season 2022-23. The test season (2023-24) has not been used.

1. Model comparison (C-index, 95% CI; difference vs Cox with paired 95% CI)

Cox baseline: 0.764 (0.725–0.799)

DeepSurv: 0.767 (0.728–0.800); vs Cox +0.002 (-0.007 to +0.013)

Transformer + DeepSurv (20 games + all features): 0.774 (0.736–0.806); vs Cox +0.010 (-0.003 to +0.024)

Transformer + DeepSurv (20 games, no workload aggregates): 0.772 (0.734–0.806); vs Cox +0.007 (-0.008 to +0.024)

Transformer + DeepSurv (5 games): 0.779 (0.741–0.816); vs Cox +0.015 (+0.003 to +0.029)

Transformer + DeepSurv (10 games): 0.776 (0.741–0.811); vs Cox +0.011 (-0.001 to +0.025)

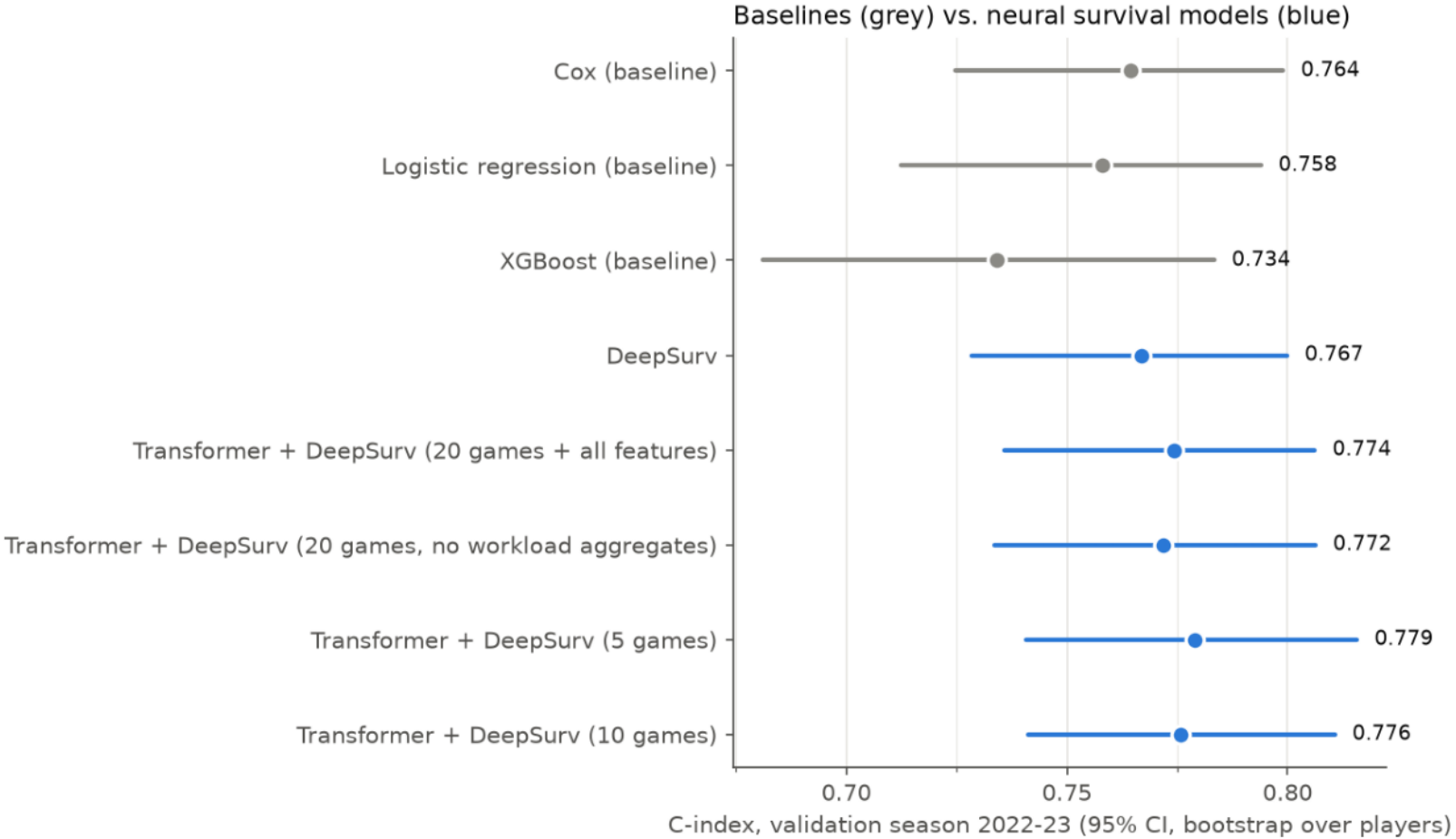
- Transformer versions vs Cox: +0.007 to +0.015 (all above Cox); 1 of 4 with a paired CI above 0.
- DeepSurv on the same summary features vs Cox: +0.002 — the gain comes from the game-by-game sequence.
- Best on validation: Transformer + DeepSurv (5 games), chosen among 5 neural models on this season; the sealed test season decides whether its advantage over Cox is real.

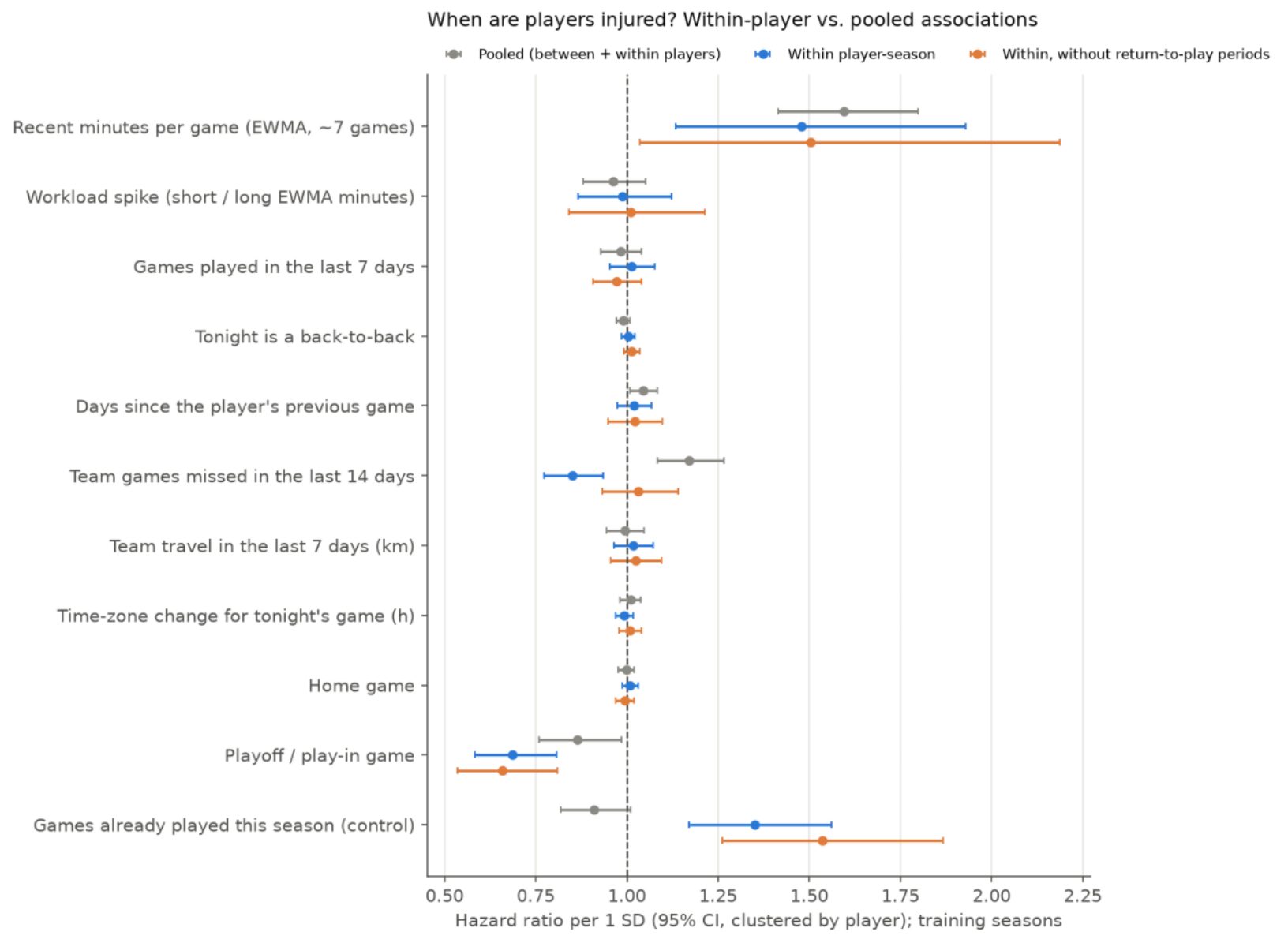
2. Timing: what precedes injury within the same player and season?

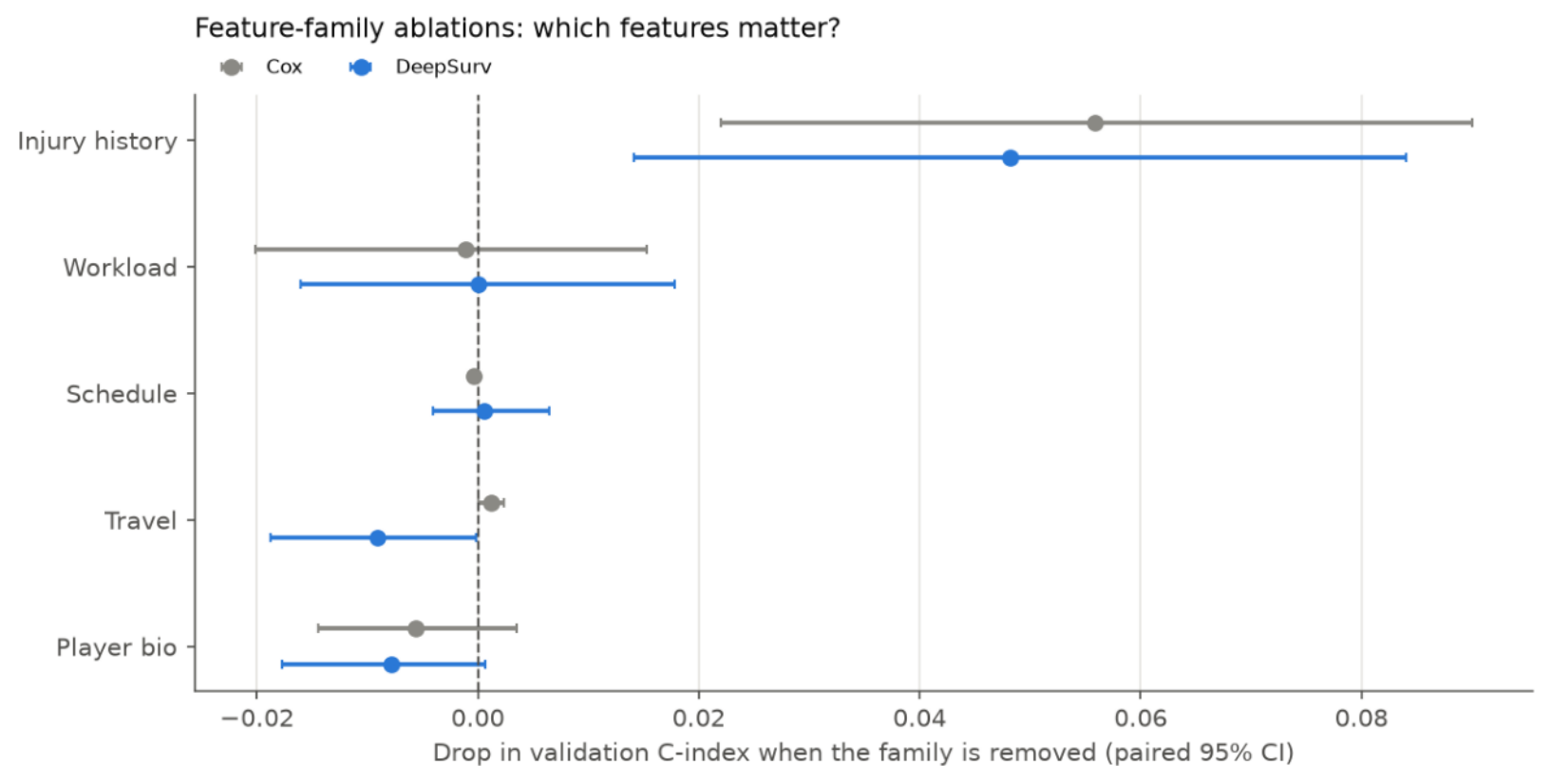
- Heavier recent minutes: HR 1.48 per SD (1.13–1.93); without return-to-play periods 1.50 (1.03–2.19).
- Workload spikes (ACWR), back-to-backs, games in 7 days, travel and time zones: no association.
- Missed games looked protective (HR 0.85) but not after removing return-to-play periods (HR 1.03): an artifact.

3. Feature-family ablations

- Injury history is the only family that reliably helps. Removing it costs:
  Cox: C −0.056 (paired 95% CI 0.022–0.090)
  DeepSurv: C −0.048 (paired 95% CI 0.014–0.084)
- Removing workload, schedule, travel or bio changes the C-index by at most 0.009.


Baselines (grey) vs. neural survival models (blue)
Cox (baseline)
Logistic regression (baseline)
XGBoost (baseline)
DeepSurv
Transformer + DeepSurv (20 games + all features)
Transformer + DeepSurv (20 games, no workload aggregates)
Transformer + DeepSurv (5 games)
Transformer + DeepSurv (10 games)
0.764
0.758
0.734
0.767
0.774
0.772
0.779
0.776
0.70
0.75
0.80
C-index, validation season 2022-23 (95% CI, bootstrap over players)


When are players injured? Within-player vs. pooled associations
Pooled (between + within players)
Within player-season
Within, without return-to-play periods
Recent minutes per game (EWMA, ~7 games)
Workload spike (short / long EWMA minutes)
Games played in the last 7 days
Tonight is a back-to-back
Days since the player's previous game
Team games missed in the last 14 days
Team travel in the last 7 days (km)
Time-zone change for tonight's game (h)
Home game
Playoff / play-in game
Games already played this season (control)
0.50
0.75
1.00
1.25
1.50
1.75
2.00
2.25
Hazard ratio per 1 SD (95% CI, clustered by player); training seasons


Feature-family ablations: which features matter?
Cox
DeepSurv
Injury history
Workload
Schedule
Travel
Player bio
−0.02
0.00
0.02
0.04
0.06
0.08
Drop in validation C-index when the family is removed (paired 95% CI)